BROKO AI PLATFORM

Workflow Architecture & Self-Learning System Blueprint v3.0

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This document defines the complete automation architecture: every workflow, every feedback loop, every self-improvement mechanism, and the AI tools that power them.

ARCHITECTURE PRINCIPLE — AUTOMATION FIRST

Every process at Broko must be designed with automation as the default, not as an afterthought. The question for every task is not "should we automate this?" — it is "when will we automate this?". Human intervention is the exception, not the rule.

THE THREE-LAYER ARCHITECTURE:

Layer 1: Capture	Layer 2: Intelligence	Layer 3: Execution
Every data point — lead behaviour, agent action, deal outcome, market change — is captured automatically. Nothing relies on human memory or manual entry.	Claude AI, ML models, RAG bots, and validation systems process captured data in real-time. Insights surface to humans instead of data.	n8n workflows act on insights automatically — send message, update CRM, retrain model, generate report, trigger alert. Humans approve strategy, not tasks.

WORKFLOW 1 — LEAD GENERATION TO AGENT

A lead enters Broko. Within 60 seconds, it is captured, scored, enriched, and routed to the right agent — with no human touching it.

1	**Lead Enters**	Ad / Website / Referral	Visitor submits form, clicks Meta/Google ad, or types into free-flow search. Consent captured at point of entry (CASL).	Meta Ads API, Google Ads API, CRM form capture
			▼ <60 seconds ▼	
2	**Deduplication + CASL Check**	n8n automation	Same phone/email from multiple sources → deduplicated to 1 record. CASL consent recorded. Lead tagged with source.	n8n, MongoDB duplicate check
			▼	
3	**AI Scoring**	Ma3 Thrilok ML Model	Behaviour data (pages, dwell time, Bestimate clicks, return visits) → score 0-100. Hot=70+, Warm=40-69, Cold=<40.	XGBoost model, Redis real-time, MLflow
			▼	
4	**Agent Assignment**	n8n + CRM Rules	Hot → senior agents by territory. Warm → standard queue by postal code. Unassigned >5 min → escalate to Ivan.	CRM assignment engine, Google Maps API postal code match
			▼ Hot leads only ▼	
5	**AI Voice Call (Hot Leads)**	Vapi.ai + ElevenLabs	Within 60 seconds of capture: AI calls lead. Greets by name, confirms interest, asks 2 qualifying questions, books agent callback.	Vapi.ai, ElevenLabs voice, Claude API conversation logic
			▼	
6	**AI Nurturing Sequence**	n8n + Wati.io	WhatsApp + email + SMS sequence triggered by behaviour. Max 1 message/day. All CASL compliant with unsubscribe.	Wati.io, Mailchimp, Twilio, Claude AI personalisation
			▼	
7	**Lead Converted / Lost**	CRM Outcome Logging	Every outcome logged as training signal. Converted: score 1.0. Lost: score 0.0. Reason logged if known.	CRM outcome module, MLflow training signal
			▼	
8	**Model Retraining**	MLflow Pipeline	Sunday night: new outcomes + new behavioural data → retrain scoring model. Better predictions next week.	MLflow, AWS SageMaker, Python XGBoost

WORKFLOW 2 — CMA REQUEST TO BRANDED REPORT

Agent enters address. 10 seconds later, a professional branded PDF report is ready to share with clients. No manual research. No manual calculations.

1	**Agent Enters Address**	Agent Portal	Agent types address in CMA module. Property specs auto-fetched from MLS (stays on AWS — never leaves).	Agent portal, MLS data API, AWS VPC
			▼ All on AWS ▼	
2	**Hard Value Model**	XGBoost (AWS SageMaker)	Property specs: sq ft, bedrooms, lot size, garage, year built → price contribution from physical attributes. AWS only.	XGBoost, AWS SageMaker, MLflow
			▼	
3	**Soft Features Model**	ML Model (AWS)	Qualitative factors: condition, renovation quality, curb appeal scoring → soft value adjustment. AWS only.	Python ML model, AWS SageMaker
			▼	
4	**Market Analysis Model**	ML Model (AWS)	Absorption rate, days on market, comparable sales, market conditions → market adjustment factor. AWS only.	Python ML model, MLS comparables, AWS
			▼ Combined price estimate ▼	
5	**AI Report Writing**	Claude API (non-sensitive only)	Summarised data (NOT raw MLS) sent to Claude. Claude writes: price narrative, market context, comparable explanation in plain English.	Claude API claude-sonnet-4-6, non-sensitive payload only
			▼ <10 seconds ▼	
6	**PDF Generation**	AWS Lambda + ReportLab	Claude narrative + ML price estimate + Broko brand template + agent photo → branded PDF. Downloadable instantly.	AWS Lambda, ReportLab, Broko template
			▼	
7	**Agent Rates CMA**	Agent Portal	Thumbs up/down + optional note. After 50 ratings → Sunday night retraining triggered automatically.	Feedback API, MLflow trigger

WORKFLOW 3 — TRANSCRIPT TO BACKLOG TO NOTION

Pankaj speaks (call, voice memo, or typed). Claude processes the transcript. Notion backlog is updated. David gets notified. All automated.

1	**Call / Meeting Happens**	Pankaj + team member	Discussion about features, bugs, ideas, compliance issues, investor feedback — any conversation.	Zoom, phone, WhatsApp — any medium
▼				
2	**Transcript Uploaded**	Pankaj	Audio transcript or text summary uploaded to Claude in Broko Enhancements project. Can be raw voice-to-text.	Claude.ai Broko Enhancements project
▼ **ai-product-backlog-creator skill activates** ▼				
3	**Skill Processes Transcript**	Claude AI (ai-product-backlog-creator skill)	Deduplicates, extracts actionable items, assigns modules, sets priorities (P0/P1/P2), identifies owners, flags discussion points.	Claude skill: ai-product-backlog-creator, Broko Enhancements project
▼				
4	**Notion-Ready CSV Generated**	Claude AI	CSV file with all 21 columns populated: Ticket ID, Module, Task, Owner, Priority, Discussion Points, Feedback Loop, Risks, Growth Opportunity, Automation Opportunity.	Claude output, 21-column Notion schema
▼				
5	**Notion Import + Notifications**	Br2 David (intern PM)	David imports CSV to Notion. n8n auto-sends WhatsApp notification to each task owner with their assigned items.	Notion API, n8n, WhatsApp Business API
▼				
6	**6-Minute Check-in**	Pankaj + task owner	Pankaj uses Discussion Points column for every 6-minute 1-on-1. Module owners give status updates. Blockers escalated.	Notion board, discussion points from roadmap

ALL SELF-LEARNING FEEDBACK LOOPS

Every output at Broko improves the next output. This is the self-improvement mandate. Below are all 10 automated feedback loops running in the platform.

LOOP 1 — CMA ACCURACY IMPROVEMENT

TRIGGER	ACTION	OUTPUT	OWNER	CADENCE
50+ agent ratings on CMA outputs (thumbs up/down + notes)	MLflow retraining pipeline triggers Sunday night → model evaluated on held-out test set → if improved, auto-deployed	More accurate CMA predictions — error reduces week on week. Target: MAPE <2% by Month 6	Ma3 Thrilok reviews weekly accuracy report	**Automatic**

LOOP 2 — LEAD SCORING IMPROVEMENT

TRIGGER	ACTION	OUTPUT	OWNER	CADENCE
New lead outcomes logged (converted / lost / in progress)	Lead scoring model retrains every Sunday night on new outcome data → scoring weights updated → predictions improve	Better lead scores → agents contact highest-intent leads first → higher conversion rate	Ma3 Thrilok monitors model drift weekly	**Automatic**

LOOP 3 — AD CREATIVE OPTIMISATION

TRIGGER	ACTION	OUTPUT	OWNER	CADENCE
Ad performance data (CTR, CPC, conversion rate) pulled via API daily	Claude analyses performance patterns weekly → generates top 3 new creative variants → Brandlume implements	Lower cost per lead over time. Winning creative patterns learned and replicated.	Br5 Deepanshu reviews weekly performance vs KPIs	**Automatic**

LOOP 4 — SEO CONTENT IMPROVEMENT

TRIGGER	ACTION	OUTPUT	OWNER	CADENCE
Google Search Console rankings + Semrush traffic data weekly	Claude identifies keywords where ranking 4-10 (low-hanging fruit) → generates targeted content brief → Megha approves → Haresh executes	Organic traffic grows compounding monthly. Top-3 rankings on high-intent Ontario real estate keywords.	Ma10 Megha reviews SEO recommendations weekly	**Automatic**

LOOP 5 — BOT KNOWLEDGE IMPROVEMENT

TRIGGER	ACTION	OUTPUT	OWNER	CADENCE
Queries that bot cannot answer (low confidence	Unanswered queries logged → Mihail provides correct answers → new Q&A pairs added to Broko	Broko Brain knowledge base grows with every escalation. Escalation rate trends down	Br3 Mihail reviews unanswered question queue	**Automatic**

	/ escalated to human)	Brain → same query answered next time	weekly.	weekly	

LOOP 6 — HOMEPAGE CONVERSION OPTIMISATION

TRIGGER	ACTION	OUTPUT	OWNER	CADENCE
Hotjar heatmaps, session recordings, A/B test results weekly	Claude analyses where users drop off → generates top 3 specific UI changes → Subhashis implements → conversion rate tracked	Homepage conversion rate improves monthly. Higher conversion = more leads from same traffic.	Ma7 Subhashis reviews Hotjar analysis weekly	**Automatic**

LOOP 7 — LEAD NURTURING OPTIMISATION

TRIGGER	ACTION	OUTPUT	OWNER	CADENCE
Email/ WhatsApp/ SMS open rates, click rates, reply rates per sequence	Underperforming sequences paused → Claude generates new variants → A/B tested against control → winner deployed	Message engagement rates improve continuously. Lead-to-showing conversion increases over time.	Ma5 Arnab monitors delivery and engagement metrics weekly	**Automatic**

LOOP 8 — AGENT PERFORMANCE COACHING

TRIGGER	ACTION	OUTPUT	OWNER	CADENCE
Agent response time, lead-to-showing rate, closing rate tracked in CRM weekly	Agents below SLA thresholds automatically flagged → Ivan receives coaching alert with specific data → targeted 1-on-1 scheduled	Underperforming agents get support before they churn. Top performers identified for recruitment referral programme.	Br4 Ivan reviews agent performance alerts weekly	**Automatic**

LOOP 9 — INFRASTRUCTURE COST REDUCTION

TRIGGER	ACTION	OUTPUT	OWNER	CADENCE
AWS Cost Explorer data pulled monthly automatically	Claude analyses cost report → identifies top 3 largest waste items → recommends specific actions → Praveen implements	Monthly AWS bill reduces over time. Target under $3,500/month Phase 1, under $8,000 Phase 2.	Ma1 Praveen reviews monthly cost report and implements top recommendations	**Automatic**

LOOP 10 — INNOVATION BOARD (CEO INTELLIGENCE)

TRIGGER	ACTION	OUTPUT	OWNER	CADENCE
Competitor website changes, Google Trends, Meta Ad Library, industry news	n8n aggregates data → Claude generates Monday 8am CEO brief with top 3 strategic priorities and action items	CEO has actionable intelligence every week without spending hours on research. Decisions made faster.	Br1 Pankaj reviews Monday morning brief (30 min max)	**Automatic**

	— pulled every Sunday				

FULL AI + TOOL STACK

Tool	Purpose	Data Sensitivity	Owner
Claude API (claude-sonnet-4-6)	CMA reports, bot conversations, compliance Q&A, offer narrative, content generation, backlog creation	🟢 **Non-sensitive only**	Ma3 Thrilok
AWS SageMaker	ML model training on proprietary sold data — CMA engine, lead scoring — stays in VPC always	🔴 **Sensitive — AWS only**	Ma3 Thrilok
XGBoost + Linear Regression	CMA three-pillar model — hard value, soft features, market analysis	🔴 **AWS only — Core IP**	Ma3 Thrilok
MLflow	ML model versioning, experiment tracking, automated retraining pipeline	🔴 **Sensitive**	Ma3 Thrilok
Pinecone (via AWS PrivateLink)	Vector store for Broko Brain RAG — all 10 bots query this before answering	🟡 **Semi-sensitive**	Ma3 Thrilok
LangChain	RAG pipeline, document retrieval, chain orchestration for bots	🟡 **Depends on payload**	Ma3 Thrilok
LangGraph	Multi-step agentic workflows — software humanoid orchestration	🟢 **Non-sensitive**	Ma2 Vikas
Langfuse (self-hosted)	AI output monitoring, hallucination detection, cost tracking — all data stays internal	🔴 **All — self-hosted**	Ma1 Praveen
n8n (self-hosted)	All workflow automation — marketing, CRM, alerts, lead nurturing, retraining triggers	🟡 **Semi-sensitive**	Ma1 Praveen
OpenAI Whisper	Audio-to-text for meeting transcripts — not used with MLS data	🟢 **Non-sensitive**	Ma1 Praveen
ElevenLabs + Vapi.ai	Voice AI for lead follow-up calls and open house bot interactions	🟢 **Non-sensitive**	Ma3 Thrilok
GitHub Copilot + Cursor	Developer AI assistance — code generation and review for all developers daily	🟢 **Code only**	All developers
Microsoft Clarity	Heatmaps and session recordings — consumer behaviour on broko.ai	🟢 **Anonymised**	Ma7 Subhashis
AWS (EC2, ECS, CloudFront, S3, SES, EventBridge, SageMaker, Lambda, GuardDuty, WAF)	Compute, storage, CDN, email, events, ML, serverless, security — full AWS stack	🔴 **All data**	Ma1 Praveen
MongoDB Atlas	Primary database — all CRM, deals, agent, consumer data	🔴 **Sensitive**	Ma5 Arnab
DocuSign / Signwell	E-signature for all transaction documents	🔴 **Transaction data**	Ma4 Debasis
Wati.io + Twilio	WhatsApp and SMS delivery for lead nurturing and alerts	🟡 **Contact data**	Ma5 Arnab
Semrush API	SEO performance data, keyword tracking, competitor monitoring	🟢 **Public data**	Ma10 Megha

Google Ads API + Meta Ads Manager	Paid advertising performance data and campaign management	🟢 **Ad data**	Ve1 Brandlume
Playwright + Pytest + k6	Automated testing — E2E, API, load testing in CI/CD pipeline	🟢 **Test data**	Ma9 Tantusree
Sentry	Error capture and monitoring in production — triggers RCA workflow	🟢 **Error logs**	Ma9 Tantusree
Linear / Jira	Bug tracking, sprint management, RCA documentation	🟢 **Internal**	Br2 David
Metabase (self-hosted)	Business intelligence dashboards for CEO and team — data stays internal	🔴 **All — self-hosted**	Ma1 Praveen
HoodQ / Walk Score / Google Places API	Neighbourhood intelligence — schools, transit, walkability for listings	🟢 **Public data**	Ma3 Thrilok
QuickBooks Online	Accounting, commission calculations, P&L, investor reporting	🔴 **Financial data**	Br3 Mihail

DATA BOUNDARY ARCHITECTURE — NON-NEGOTIABLE

This is the most important technical compliance requirement. Failure to enforce this puts our PropTx data licence at risk — which would take down the entire platform.

🔴 STAYS ON AWS — NEVER LEAVES	🟢 CAN GO TO CLAUDE API
• Raw MLS listing data (addresses, prices, dates) • Individual sold transaction data • Agent commission information • Client personal information (PII) • Property-level financial data • RECO member data • FINTRAC identification data • Any data identifying specific buyers, sellers, or agents	• Anonymised market trend summaries (e.g. "average days on market in Mississauga increased 12%") • CMA price estimate (single number, no raw data) • RECO compliance questions (no deal-specific data) • Compliance Q&A (general RECO knowledge) • Marketing copy and content generation • Offer document narrative (agent-reviewed before sending) • Training content and FAQ answers • Social media and marketing material generation

ENFORCEMENT MECHANISM:

- AWS API Gateway — all external API calls logged and audited. Any call containing prohibited data patterns is blocked.
- Monthly data boundary audit by Ma1 Praveen — review all Claude API call logs for any prohibited data patterns.
- Automated test in CI/CD pipeline — checks that no feature sends raw MLS data to any external API endpoint.
- Ma3 Thrilok maintains explicit whitelist of what can be sent to Claude API — all developers must follow it.

WEEKLY AUTOMATION CALENDAR

This is what runs automatically every week — no human intervention required.

Time	What Runs	Output	Owner notified
Sunday 11pm	CMA model retrain pipeline (MLflow)	Updated model deployed if accuracy improves	Ma3 Thrilok
Sunday 11:30pm	Lead scoring model retrain pipeline	Updated lead scoring weights deployed	Ma3 Thrilok
Monday 7am	Competitor monitoring scrape (Semrush + Google + Meta Ad Library)	Competitor intel data ready for Innovation Bot	n8n automated
Monday 8am	Innovation Board Bot generates CEO brief	1-page strategy brief delivered to Pankaj via WhatsApp	Br1 Pankaj reads
Monday 8am	7-day social content calendar generated by Claude	Caption batch + posting schedule sent to Haresh	Ma11 Haresh executes
Monday 8:30am	Sprint standup reminder sent to all module owners	WhatsApp message: "Standup in 30 min — bring your update"	All module owners
Wednesday 6am	Brandlume ad performance report pulled via API	Performance dashboard updated — alerts if KPI below threshold	Br5 Deepanshu reviews
Friday 9am	AWS cost report generated from Cost Explorer	Cost vs budget comparison with top 3 recommendations	Ma1 Praveen reviews
Friday 4pm	Weekly sprint retrospective summary	What shipped vs planned — exported to Notion	Br2 David updates board
1st of month 7am	Investor update drafted by Claude from CRM + financial data	Draft email ready for Pankaj review — GCI, agents, milestones	Br1 Pankaj sends
Daily 9am	Price alert triggers processed from overnight MLS changes	Alerts sent to all subscribed agents and clients	Automated delivery
Daily 6am	Neighbourhood SEO page data refreshed from MLS API	All 500+ pages show current pricing and market data	Automated — no owner

"The system improves itself. Humans improve the strategy."

— Broko AI Platform Architecture Principle